Calculus Skeleton Notes

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Average Rates of Change

Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits
Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

b) Sketch a possible graph

c) How could I approximate my speed at $t = 10$ minutes?

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2).

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1. We need to investigate and define this limiting process more precisely. This DESMOS visualizes the idea: `https://www.desmos.com/calculator/oi7poyhodr`

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace

1.1 - Limits

Goals: Describe limits graphically, numerically, and algebraically.
Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \qquad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x+1)(x-1)}{x-1}, \qquad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?

Definition:
$\lim_{x \to a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a.

Example 2: Compute the limit as $x \to 1$ of $f(x)$, $g(x)$, and $h(x)$.

Example 3: Use a table of values to estimate $\lim_{x \to 1}(x + 1)$.

Example 4: Use a table of values to estimate $\lim_{x \to 0} \frac{1}{x}$.

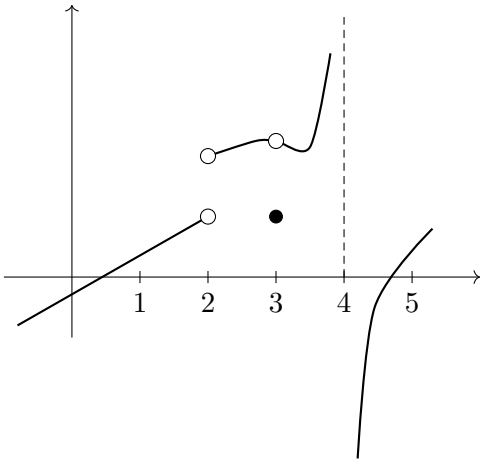
Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

Definition:

$\lim_{x \to a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right, i.e. $x > a$.

Example 6: Use the graph to evaluate the indicated limits.



$\lim_{x\to 2^-} f(x) =$ _____, $\lim_{x\to 2^+} f(x) =$ _____, $\lim_{x\to 2} f(x) =$ _____,

$\lim_{x\to 3^-} f(x) =$ _____, $\lim_{x\to 3^+} f(x) =$ _____, $\lim_{x\to 3} f(x) =$ _____,

$\lim_{x\to 4^-} f(x) =$ _____, $\lim_{x\to 4^+} f(x) =$ _____, $\lim_{x\to 4} f(x) =$ _____.

Example 7: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$

Example 8: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15

1.3 - Finding Limits Analytically

Goals: Evaluate limits using limit laws, algebra, and the Squeeze Theorem.
Before Class Videos: 1.3.1–1.3.3

Definition:

If $f(x)$ gets arbitrarily close to L for all x sufficiently close to a, then

$$\lim_{x \to a} f(x) = L.$$

For many "nice" functions, evaluating a limit just means plugging in the number. We'll investigate what functions are "nice" in section 1.5 on Continuity. For example:

Example 1:

- $\lim_{x \to 2} x^3 =$
- $\lim_{x \to \pi} \sin(x) =$
- $\lim_{x \to e} \ln(x) =$
- $\lim_{x \to 4} \sqrt{x} =$

Theorem:

If L, M, c, and k are real numbers and

$$\lim_{x \to c} f(x) = L \qquad \text{and} \qquad \lim_{x \to c} g(x) = M,$$

then

1. Sum Rule:	$\lim_{x \to c}[f(x) + g(x)] = L + M$
2. Difference Rule:	$\lim_{x \to c}[f(x) - g(x)] = L - M$
3. Constant Multiple Rule:	$\lim_{x \to c}[kf(x)] = kL$
4. Product Rule:	$\lim_{x \to c}[f(x)g(x)] = LM$
5. Quotient Rule:	$\lim_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \quad M \neq 0$
6. Power Rule:	$\lim_{x \to c}[f(x)]^n = L^n, \quad n$ a positive integer
7. Root Rule:	$\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n$ a positive integer

If n is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing c.

Example 2: Evaluate

$$\lim_{x\to 2} \frac{x^2 - 2x}{x + 2}.$$

Example 3: Come up with functions f and g such that

$$\lim_{x\to 3} f(x)g(x) \neq \left[\lim_{x\to 3} f(x)\right] \left[\lim_{x\to 3} g(x)\right].$$

Four Algebraic Tricks

Example 4:

$$\lim_{x\to 2} \frac{x^2 - 2x}{x - 2}.$$

Example 5:

$$\lim_{x\to 0} \frac{(x-2)^2 - 4}{x}.$$

Example 6:

$$\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9}.$$

Example 7:

$$\lim_{x\to 2} \frac{\frac{1}{2}-\frac{1}{x}}{x-2}.$$

Example 8: Evaluate

$$\lim_{x\to 4} \frac{4-x}{5-\sqrt{x^2+9}}.$$

Theorem:

Squeeze Theorem. If $f(x) \leq g(x) \leq h(x)$ for all x in an interval containing a (except possibly at a), and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L,$$

then

$$\lim_{x \to a} g(x) = L.$$

Example 9: Evaluate

$$\lim_{x \to 0} x^2 \sin(1/x^2).$$

Example 10: Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all x. For what value of a can you determine $\lim_{x \to a} g(x)$? What is the limit?

Suggested after class problems from APEX: Section 1.3 problems 8,11,29,36

1.5 - Continuity

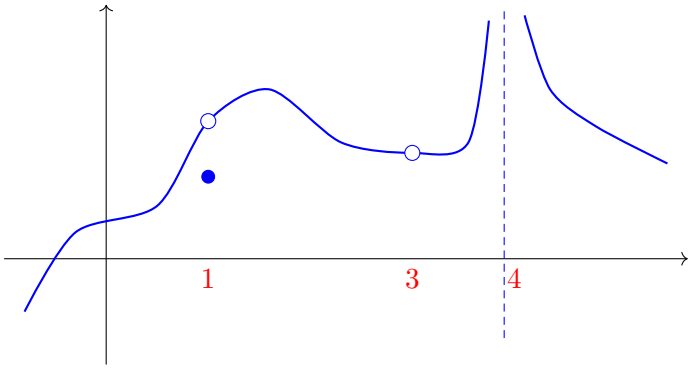
Goals: Determine where a function is continuous, build continuous piecewise functions, and apply the Intermediate Value Theorem.
Before Class Videos: Section 1.5 videos 1 and 3

Definition:

A function f is **continuous at** c if all three of the following hold:

1. $f(c)$ exists,
2. $\lim_{x \to c} f(x)$ exists, and
3. $\lim_{x \to c} f(x) = f(c)$.

Example 1: Identify the type of discontinuity at each marked x-value.



Example 2: For what values of a is

$$f(x) = \begin{cases} x + 2, & x \leq 1, \\ ax^2, & x > 1 \end{cases}$$

continuous everywhere?

Example 3: For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous everywhere?

Combining Continuous Functions

Theorem:

If f and g are continuous at c, then $f+g$, $f-g$, constant multiples of f, and fg are continuous at c. The quotient f/g is also continuous at c provided $g(c) \neq 0$.

Let's justify the sum rule from the limit laws:

Composition of Functions

Example 4: Compute the domain and range of $\sin\sqrt{x}$

Theorem:

Suppose f is continuous at a and g is continuous at $f(a)$. Then the composition $g \circ f$ is continuous at a, and

$$\lim_{x \to a} g(f(x)) = g(f(a)).$$

Example 5: On what intervals is $\sin\left(\frac{x-2}{x^2+5x+7}\right)$ continuous?

Example 6: Evaluate

$$\lim_{x \to 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2}.$$

Theorem:

Intermediate Value Theorem. If f is continuous on $[a, b]$ and y_0 is between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that

$$f(c) = y_0.$$

Sketch:

Example 7: Sketch examples where the assumptions of the Intermediate Value Theorem are not satisfied.

Example 8: Show that

$$x^3 - 3x - 1 = 0$$

has a solution in $[0, 2]$.

Example 9: Use the intermediate value theorem to show that at some time a baby's height equalled their child's height.

Suggested after class problems from APEX: Section 1.5 problems 21,24,35

1.6 - Limits Involving Infinity

Goals: Investigate the special $\sin(x)/x$ limit, compute limits whose values are infinity, as well as limits at infinity

Before Class Videos: 1.6.1-1.6.3

Example 1: Explore $\lim_{x \to 0} \dfrac{\sin x}{x}$.

Example 2: Compute $\lim_{x \to 0} \dfrac{\sin(5x)}{\sin(3x)}$.

Example 3: Explore

$$f(x) = \frac{x}{x-2}$$

and find all asymptotes.

Definition:

$\lim_{x \to a} f(x) = \infty$ if $f(x)$ can be made arbitrarily large for all values of x sufficiently close to a.
Note: A limit being infinite is a particular example of a limit not existing.

Definition:

$\lim_{x \to \infty} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently large.

Example 4:

$$\lim_{x \to \infty} \sin(1/x)$$

Example 5:

$$\lim_{x \to \infty} x \sin(1/x)$$

Example 6: Find all Horizontal Asymptotes for

$$\frac{2x^3 + 3x}{x^4 - x^2}.$$

Example 7: Find all Horizontal Asymptotes for

$$\frac{\sqrt{x^6+2x}}{x^3+x^{1/3}}.$$

Example 8: Find VA's; HA for $e^{1/x}$; graph it!

Example 9: Give example of a function with two HAs, $y=0$ and $y=2$, continuous everywhere. Hint: $\arctan(x)$.

Suggested after class problems from APEX: Section 1.4 #40 and Section 1.6 #10, 19,27

2.1 - Derivatives

Goals: Define the Derivative and apply it to initial examples
Before Class Videos: 2.1.1, 2.1.2

Example 1: Recall the velocity problem from the first day of class. If Trefor has some distance function $f(t)$, how fast is he going at 10 minutes? Let's explore graphically and numerically:

Definition:
The derivative of $f(x)$ at c is given by

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say f is differentiable at c.

Example 2: Compute the derivative of $f(x) = x^2$ at $x = 2$.

Definition:
The derivative of $f(x)$ on an open interval I is the function $f'(x)$ given by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists for all x in I. If it does we say $f(x)$ is differentiable on I.

Example 3: $f(x) = 1/x^2$.

Example 4: $f(x) = \sqrt{x}$.

Example 5: Let's explore if the following functions are differentiable at $x = 0$:

a)

$$f(x) = \begin{cases} 2, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

b) $f(x) = |x|$

c) $f(x) = x^{2/3}$

d) $f(x) = x^{1/3}$

Suggested after class problems from APEX: Section 2.1: 9, 17, 24

2.2 - Derivatives

Goals: A shorter than normal worksheet with a few extras on derivatives
Before Class Videos: Optional tricky example from last class's content: 2.1.4

Definition:
The derivative of $f(x)$ at c is given by

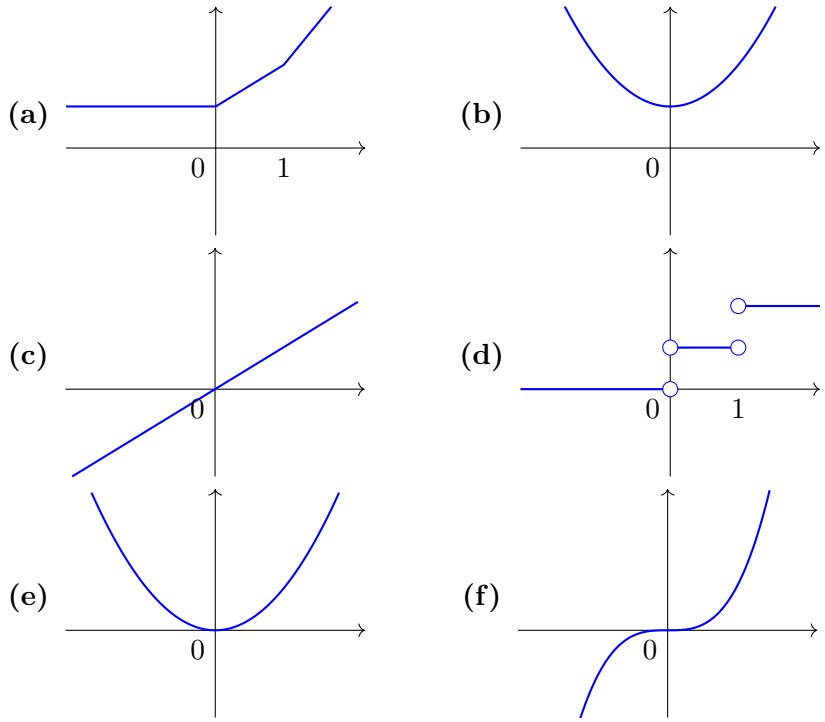
$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say f is differentiable at c.

Definition:
An alternate formula for the derivative of $f(x)$ at c is

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

Example 1: The following six graphs consists of three pairs of functions and their derivative. Match them up.



Theorem:
Differentiable $\implies$ Continuous.

Note: This worksheet is deliberately short. I anticipate this class being a bit behind and finishing up prior material. If I am wrong I will edit it and add to it. The next worksheet is a more normal length.

Suggested after class problems from APEX: Section 2.2 15,17

2.3 - Differentiation Rules

Goals: State and apply Derivative Rules
Before Class Videos: 2.3.1

Theorem (Power Rule):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n \neq 0$

Theorem:

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

1. **Sum/Difference Rule**:

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

2. **Constant Multiple Rule**:

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

Example 1: Compute the derivative of $f(x) = 3x^4 - 5\sqrt{x} + 6x^{3/2} - \frac{7}{x^2} + 1$.

Proof of Theorem:

Definition:
The Point-Slope formula for the equation of the straight line through (x_0, y_0) with slope m is

$$y = y_0 + m(x - x_0)$$

Example 2: Find the equation of tangent line to a differentiable function $f(x)$ at the point x_0.

Example 3: Find the equation of the tangent line to the function $f(x) = 2x^2 + 1$ at the point $(1, 3)$.

Common Derivatives:

Example 4: Explore the graphs of the following functions to discover their derivatives.

- $\frac{d}{dx}(e^x) =$

- $\frac{d}{dx}\ln(x) =$ for $x > 0$

- $\frac{d}{dx}\sin(x) =$

- $\frac{d}{dx}\cos(x) =$

Higher Derivatives: The derivative $f'(x)$ is itself a function we can take the derivative of. We call this the second derivative and might denote it by

$$f''(x) \qquad \text{or} \qquad \frac{d^2}{dx^2}f(x).$$

Example 5: Compute $f''(x)$ when $f(x) = e^x + \ln(x)$.

Example 6: If $f(x) = \sin(x)$ compute

- $f'(x) =$
- $f''(x) =$
- $f'''(x) =$
- $f^{(4)}(x) =$
- $f^{(2026)}(x) =$

Example 7: If $d(t)$ represents the position of a particle as a function of time, interpret $d'(t)$ and $d''(t)$.

Example 8: If $v(t)$ represents Trefor's total YouTube views after t days, interpret $v'(t)$ and $v''(t)$.

Suggested after class problems from APEX: Section 2.3 16, 21, 34 (tangent only)

2.4 - Product and Quotient Rule

Goals: Compute derivatives using the product and quotient Rules, including of trigonometric functions

Before Class Videos: 2.4.1 and 2.4.2

Theorem (Derivative Rules):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n > 0$ is an integer.

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

- $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$
- $\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$

Let's add two new rules to the list

Theorem (Product and Quotient Rules):

$$\frac{d}{dx}[fg] = f'g + fg'$$

$$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - fg'}{g^2}$$

Example 1: Compute the following derivatives

a)

$$\sqrt{x}e^x$$

b)

$$\frac{\sin(x)}{x^3}$$

c)

$$\frac{xe^x}{x^2+1}$$

Example 2: Let's explore the intution behind why the product rule is NOT just the product of the derivatives.

Example 3: Let's finally figure out what exactly e^x is all about. We'll use the definition of the derivative to compute, for $a > 0$,

$$\frac{d}{dx}a^x$$

Let's explore

$$\lim_{h\to 0} \frac{a^h - 1}{h}$$

Definition:

The number e is defined to be the value of a such that $\lim_{h\to 0} \frac{a^h - 1}{h}$

Right-Triangle Definitions

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}} \quad \tan x = \frac{\text{opposite}}{\text{adjacent}} \quad \sec x = \frac{\text{hypotenuse}}{\text{adjacent}}$$

$$\cos x = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \cot x = \frac{\text{adjacent}}{\text{opposite}} \quad \csc x = \frac{\text{hypotenuse}}{\text{opposite}}$$

Rewriting with Sine and Cosine

$$\sin x = \sin x \quad \tan x = \frac{\sin x}{\cos x} \quad \sec x = \frac{1}{\cos x}$$

$$\cos x = \cos x \quad \cot x = \frac{\cos x}{\sin x} \quad \csc x = \frac{1}{\sin x}$$

Example 4: Let's explore the graphs of these functions.

Example 5: Compute $\frac{d}{dx}\tan(x)$

Example 6: Compute $\frac{d}{dx}\sec(x)$

Trigonometric Derivatives

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\tan x = \sec^2 x \qquad \frac{d}{dx}\sec x = \sec x \tan x$$

$$\frac{d}{dx}\cos x = -\sin x \qquad \frac{d}{dx}\cot x = -\csc^2 x \qquad \frac{d}{dx}\csc x = -\csc x \cot x$$

Suggested after class problems from APEX: Section 2.4 7, 18, 26

2.5 - Chain Rule

Goals: Compute the derivative of compositions of functions
Before Class Videos: 2.5.1, 2.5.2

Example 1: Consider $\sin(x^2)$. This is an example of a composition $f(g(x))$ or alternatively $f \circ g$.

Example 2: $f(x) = x \sin x$, $g(x) = \cos x + 1$. Write out:

- $f(g(x)) =$
- $g(f(x)) =$
- $f(f(x)) =$
- $g(g(x)) =$

Theorem (Chain Rule):
If g differentiable at x and f differentiable at $g(x)$ then

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

Example 3: $\dfrac{d}{dx} \sin(x^2 + 1)$

Example 4: $\dfrac{d}{dx} \sin^{30}(x)$

Example 5: $\dfrac{d}{dx} \cos(3e^x)$

Example 6: $\dfrac{d}{dx}\dfrac{1}{2x+1}$

Example 7: $\dfrac{d}{dx}\cos^2(x^3)$

Example 8: $\dfrac{d}{dx}\tan(x^2e^x)$.

Leibniz Notation: let $u = g(x)$,

$$\frac{d}{dx}[f(g(x))] = \frac{df}{du}\frac{du}{dx}.$$

Example 9: Car uses $6L/100km$. Cost is $1.5\$/L$. What is the rate of change in cost per $100km$?

Example 10: Gravel is dumped from a conveyor to form pile with

$$V(h) = \frac{\pi}{12}h^3.$$

At $5m$ tall, the pile is growing by 0.1 m/s. How fast is volume growing?

Example 11: Contrast graphically the slope of $f(x)$ and $f(2x)$ for an arbitrary function $f(x)$.

Example 12: Derive the Quotient Rule.

Suggested after class problems from APEX: Section 2.5: 7,17,24

2.6 - Implicit Differentiation

Goals: Compute derivatives of functions define implicitly
Before Class Videos: 2.6.1

Example 1: Explore $x^2 + y^2 = 1$. Where does it have vertical and horizontal tangents?

Big Idea: Treat $y = y(x)$ when x is independent and y is dependent.

Example 2: Find $\dfrac{dy}{dx}$ implicitly for $\sin(xy) + x = y^2 \cos x$.

Example 3: Compute $\dfrac{d^2y}{dx^2}$ for $x^2 + y^2 = 4$.

Example 4: Now explore $x^3 + y^3 - 9xy = 0$. Find all vertical and horizontal tangents.

Logarithmic Differentiation: Recall we previously learned that $\frac{d}{dx}ln(x) = \frac{1}{x}$.

Example 5: $y = x^x$.

Example 6:

$$y = \frac{(x+2)^3(x-1)^{1/3}}{(x+7)^2}.$$

Example 7:

$$y = x^{\cos x} \cdot (\cos x)^x.$$

Suggested after class problems from APEX: Section 2.6: 15,17,40

2.7 - Inverses and Inverse Trig

Goals: Compute the Derivatives of Inverse functions and Inverse Trig functions.
Before Class Videos: 2.7.1

Note: This worksheet combined some content from inverses in general and trignometric inverses. The former was covered on Oct 7th by Josh Manzer using his own worksheet and will thus not be covered extensively by Trefor, who will focus on the inverse trig for Friday's Class.

Definition:
For a function $f(x)$, let's define $f^{-1}(x)$ so

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x.$$

Example 1: $y = 2x + 1$

Example 2: $y = x^2$

Example 3: $\dfrac{d}{dx} \ln x$

Example 4: $\dfrac{d}{dx}a^x$

Example 5: Let's derive a general formula for the inverse of a function.

Example 6: $\tan^{-1}(\tan x) = x = \tan(\tan^{-1} x)$.Try evaluating these at $x = \pi$.

Example 7: $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Example 8: $\sin^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

Example 9: Plot the graph of $\arcsin(x)$ by comparing to the graph of $\sin(x)$.

Example 10: Compute $\dfrac{d}{dx}\arcsin(x)$.

Inverse Trigonometric Functions

Function	Derivative	Domain	Range
$\arcsin x$	$\dfrac{1}{\sqrt{1-x^2}}$	$[-1,1]$	$[-\pi/2,\pi/2]$
$\arccos x$	$-\dfrac{1}{\sqrt{1-x^2}}$	$[-1,1]$	$[0,\pi]$
$\arctan x$	$\dfrac{1}{1+x^2}$	$\mathbb{R}$	$(-\pi/2,\pi/2)$
$\operatorname{arccot} x$	$-\dfrac{1}{1+x^2}$	$\mathbb{R}$	$(0,\pi)$
$\operatorname{arcsec} x$	$\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}}$	$(-\infty,-1]\cup[1,\infty)$	$[0,\pi/2)\cup(\pi/2,\pi]$
$\operatorname{arccsc} x$	$-\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}}$	$(-\infty,-1]\cup[1,\infty)$	$[-\pi/2,0)\cup(0,\pi/2]$

The derivatives of $\arcsin x$ and $\arccos x$ hold for $|x|<1$; those of $\operatorname{arcsec} x$ and $\operatorname{arccsc} x$ hold for $|x|>1$.

Example 11: $\dfrac{d}{dx}\sin^{-1}(3x^2+1)$

Example 12: Let's do one more derivativion, this time of $y = \sec^{-1}(x)$.

Example 13: $\dfrac{d}{dx}[\ln(x^2+1) + x\tan^{-1}(2x)]$

Example 14: For $\tan^{-1} x + \sin^{-1} y = \frac{\pi}{4}$ find $\left.\dfrac{dy}{dx}\right|_{(0,\frac{1}{\sqrt{2}})}$

Suggested after class problems from APEX: Section 2.7: 9,13,16,23,29

4.2 - Related Rates

Goals: Relate changing quantities by differentiating equations with respect to time.
Before Class Videos: 4.2.1

Example 1: An inverted conical tank has radius 6 ft and height 8 ft. Water is pumped in at 4 ft^3/min. How fast is the water level rising when the water is 4 ft deep?

Example 2: A rocket rises vertically at 600 m/s. An observer stands 4000 m from the launch point. How fast is the angle of elevation changing when the rocket is 3000 m high?

Method:

Example 3: Two boats are 400 km apart horizontally. Boat A goes north at 30km/h while boat B goes south at 45 km/h. How fast is the distance between them changing after 2 hours?

Example 4: A point moves along the curve $y = x^2$, with $x > 0$. Find the rate of change of its distance from $(3, 0)$ in terms of x and dx/dt.

Example 5: Suppose a bar of length d leans against a wall and slides along the ground. If $x'(t)$, the rate of change of its horizontal position is constant, is $y'(t)$ constant?

Example 6: Suppose a $2m$ tall skateboarder is a distance x from a $5m$ tall wall which has a light at the top. If $x'(t) = 2m/s$, how fast is the length of the shadow cast by the light increasing?

Example 7: A trough is 20 m long, with right triangular ends 5 m high and 5 m wide. Water flows in at 100 m^3/h. How fast is the water level rising when the water is 2 m deep?

Suggested after class problems from APEX: Section 4.2 problems 3,5,7

3.1 - Extreme Values

Goals: Identify and find absolute and relative extrema, particularly on closed intervals.
Before Class Videos: 3.1.1, 3.1.2

Definition:

Let f be defined on an interval I containing c.

1. $f(c)$ is the **absolute** (or global) maximum if $f(c) \geq f(x)$ for all x in I.
2. $f(c)$ is the **absolute** (or global) minimum if $f(c) \leq f(x)$ for all x in I.
3. $f(c)$ is a **relative** (or local) maximum if $f(c) \geq f(x)$ for all x in some interval about c within I.
4. $f(c)$ is a **relative** (or local) minimum if $f(c) \leq f(x)$ for all x in some interval about c within I.

Theorem (Extreme Value Theorem):

On a closed interval $[a, b]$, a continuous function must attain an absolute maximum and absolute minimum in the interval.

Example 1: Sketch examples where the assumption is not satisfied and the conclusion of the Extreme Value Theorem is false.

Example 2: What are the maximum and minimum values of $f(x) = x^2 - 6x + 7$ on $[2, 5]$.

Definition:
A **critical number** of f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist. A **critical point** is a pair $(c, f(c))$ when c is a critical number.

Example 3: Find the absolute maximum and minimum of $f(x) = x^{2/3}$ on $[-1, 1]$.

Extreme Value Method for Closed Intervals:

1. Find all critical numbers: $f'(x) = 0$ or DNE.
2. Evaluate $f(x)$ at each critical number (in the interval) and endpoints of the interval
3. Pick biggest/smallest.

Example 4: Find the absolute max and min of $f(x) = x^4 - 8x^2$ on $[-1, 3]$.

Example 5: $f(x) = x^{1/3}$ on $[-1, 1]$.

Theorem:
If f has a max at c and $f'(c)$ exists, then $f'(c) = 0$.

Example 6: $f(x) = \ln(x+1) + \tan x$ on $(-1, 1]$.

Suggested after class problems from APEX: Section 3.1: 7,10,17,19

3.2 - Mean Value Theorem

Goals: State and Apply Rolle's Theorem and the Mean Value Theorem
Before Class Videos: 3.2.1

Note: This was originally posted for last friday, but we never began it so it is relabelled as Oct 20th notes

Theorem (Rolle's Theorem):
If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ so $f'(c) = 0$.

Proof:

Theorem (Mean Value Theorem):
Suppose $f(x)$ is continuous on $[a, b]$ anddifferentiable on (a, b).
Then there exists c in (a, b) so

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

Example 1: $y = \sqrt{x-2}$ on $[2, 6]$.

Example 2: Suppose $f(1) = 10$ and $f'(x) \geq 2$ on $1 \leq x \leq 4$. What is a minimum possible value of $f(4)$?

Example 3: $f(x) = x^3 + 3x + 1$. Show exactly one root.

Theorem:
Suppose $f'(x) = 0$ on an interval I. Then $f(x)$ is constant.

Corollary: Suppose $f'(x) = g'(x)$ on (a, b). Then $f(x) = g(x) + C$ for some constant.

Example 4: If $f'(x) = x^2$, what is $f(x)$?

Suggested after class problems from APEX: Section 3.2: 1,5,11

3.3 - Increasing Functions

Goals: Connect properties of the derivative to the idea of increasing functions and extrema
Before Class Videos: No video for this section; consult Section 3.3 of APEX Calculus to review.

Definition:
$f(x)$ increasing on interval I if for all $a < b$ in I $f(a) < f(b)$ in I.

Definition:
$f(x)$ decreasing on interval I if for all $a < b$ in I $f(a) > f(b)$ in I.

Theorem:
If $f'(x) > 0$ on I, then $f(x)$ increasing on I.

Example 1: Decide the intervals on which $f(x) = 2x^3 - 3x^2 - 12x$ is increasing and decreasing. Sketch a very loose graph of the function.

Theorem (1st Derivative Test):

Let c be a critical number of f, and f both continuous at c and differentiable near c.

1. $f'(x)$ changes sign from $+$ to $-$ then $f(c)$ is a relative max.
2. $f'(x)$ changes sign from $-$ to $+$ then $f(c)$ is a relative min.
3. If there is no change in sign, then $f(c)$ is not a relative extremum

Example 2: Classify all critical numbers of

$$f'(x) = \frac{e^{-2x}}{20}(x-1)(x+1)^2.$$

Example 3: Classify all critical numbers of

$$g'(x) = \frac{(x+4)(x-1)^2}{x-2},$$

Example 4: Sketch graph of function such that:

- V.A. at $x = 1$;
- $f'(x) < 0$ on $(1, \infty)$;
- $f'(x) > 0$ on $(-1, 1)$;
- minimum at $x = -1$;
- $\lim_{x \to -\infty} f(x) = 0$;
- continuous elsewhere.

Example 5: Determine intervals where $f(x) = \cos x + \sin x$ is increasing and decreasingon $[0, 2\pi]$.

Suggested after class problems from APEX: Section 3.3: 15,21

3.4 - Concavity

Goals: Connect the idea of concavity to the second derivative and use this to find max and mins
Before Class Videos: 3.4.1

Definition:

A differentiable function $f(x)$ is

- concave up on an interval I if $f'(x)$ is increasing on I;
- concave down on an interval I if $f'(x)$ is decreasing on I.

Test for Concavity: If $f(x)$ is twice differentiable on an interval I,

- If $f''(x) > 0$ on I, then f is concave up on I;
- If $f''(x) < 0$ on I, then f is concave down on I.

Example 1: $y = x^3$:

Example 2:

$$f(x) = x^3 - 6x^2 + 15.$$

Theorem (2nd derivative test):

If f is continuous near c,

- if $f'(c) = 0$ and $f''(c) > 0$, f has a local min at c;
- if $f'(c) = 0$ and $f''(c) < 0$, f has a local max at c.

Definition:
A point $(c, f(c))$ on the graph of f is an inflection point if the concavity of f changes at c

Theorem:
If $(c, f(c))$ is an inflection point, then either $f''(c) = 0$ or $f''(c)$ does not exist

Example 3: $x^{1/3}$

Example 4: x^4

Example 5: Find all critical point and classify them for $3x^4 - 8x^3 - 90x^2$. Additionally find all inflection points.

Suggested after class problems from APEX: Section 3.4: 2,12,16

3.5 - Curve Sketching

Goals: Let's put everything together and now graph functions based on all the information we know

Before Class Videos: No video for this section; consult Section 3.5 of APEX Calculus to review.

Example 1: Graph

$$f(x) = \frac{x}{x-5}, \qquad f'(x) = -\frac{5}{(x-5)^2}, \qquad f''(x) = \frac{10}{(x-5)^3}.$$

- Domain:

- Intercepts:

- Critical points:

- Vertical Asymptotes:

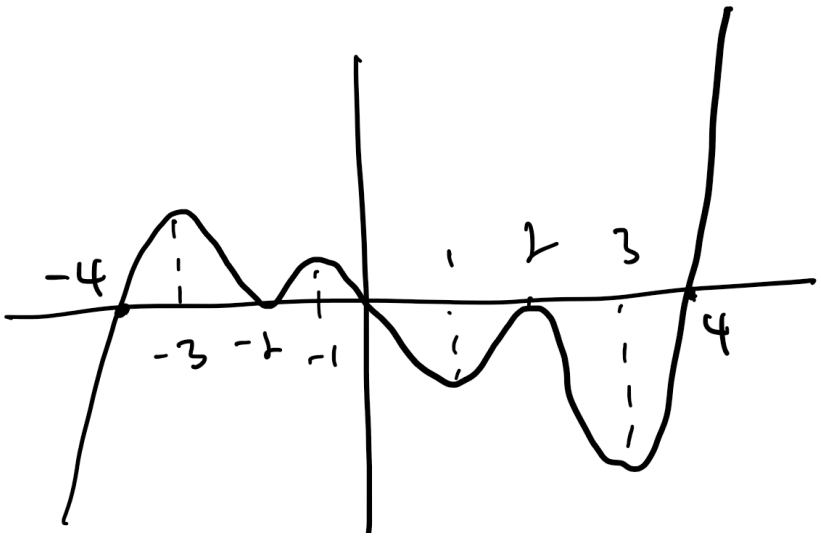
- Horizontal Asymptotes:

- Increasing/Decreasing:

- Concave Up/Down:

Let's graph it:

Example 2: Consider this plot of the derivative $f'(x)$ and classify all Critical Numbers



Example 3:

$$f(x) = \frac{1}{1+e^{-x}}, \qquad f'(x) = \frac{e^{-x}}{(1+e^{-x})^2}, \qquad f''(x) = \frac{e^{-x}(e^{-x}-1)}{(e^{-x}+1)^3}.$$

- Domain:

- Intercepts:

- Critical points:

- Vertical Asymptotes:

- Horizontal Asymptotes:

- Increasing/Decreasing:

- Concave Up/Down:

Example 4: Graph

$$f = e^{1/x}, \qquad f'(x) = -\frac{e^{1/x}}{x^2}, \qquad f''(x) = \frac{e^{1/x}(2x+1)}{x^4}.$$

- Domain:

- Intercepts:

- Critical points:

- Vertical Asymptotes:

- Horizontal Asymptotes:

- Increasing/Decreasing:

- Concave Up/Down:

Let's plot it:

Example 5: (Time permitting) Graph

$$\frac{x-x^2}{x^2-9x+8}, \qquad f'(x)=\frac{x-4}{x^2-10x+25}, \qquad f''(x)=\frac{8}{(x-5)^3}$$

Suggested after class problems from APEX: Section 3.5: 11,15,21

6.7 - L'Hopital's Rule

Goals: Recognize different types of indeterminate forms and limits using L'Hospital's Rule and
Before Class Videos: 4.5.1, 4.5.2

Which of the following are indeterminate forms?

$\frac{\infty}{\infty}$	$\frac{\infty}{0}$	$\frac{0}{\infty}$
$\frac{0}{0}$	$0 \cdot \infty$	$\infty \cdot \infty$
$\infty + \infty$	$\infty - \infty$	0^0
1^∞	0^∞	∞^0

Case 1: Quotient Forms

Theorem (L'Hospital's Rule):
If f and g are differentiable functions on an open interval I containing a and $g'(x) \neq 0$ on I except possibly at c, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.$$

Note: a can be $\pm\infty$ too, and the can be $\pm\infty$ too, and the limits can be one sided

Example 1:

$$\lim_{x \to 0} \frac{\sin x}{x}$$

Example 2:

$$\lim_{x \to 0} \frac{e^x - x - 1}{x^2}$$

Example 3:

$$\lim_{x \to 0^+} \frac{\ln(e^x - 1)}{\ln x}$$

Case 2: Exponential Forms.

1. Instead of $f(x)^{g(x)}$, study $\ln(f(x)^{g(x)})$.
2. Apply the log rule to get $g(x)\ln(f(x))$.
3. Rewrite as a quotient.
4. Apply Case 1 L'Hospital's Rule.
5. Take the exponential of the final value.

Example 4:

$$\lim_{x \to \infty} x^{e^{-x}}$$

Example 5:

$$\lim_{x \to e^+} (\ln x)^{1/(x-e)}$$

Example 6:

$$\lim_{x \to (\pi/2)^-} (\tan x)^{\cos x}$$

Case 3: Products and Differences like $\infty \cdot 0$, $\infty - \infty$, etc.

Use algebra to convert to a $0/0$ or ∞/∞ form, then apply Case 1.

Example 7:

$$\lim_{x \to (\pi/2)^-} \left(\frac{\pi}{2} - x\right) \tan x$$

Example 8:

$$\lim_{x \to 0^+} \left(\frac{3x+1}{x} - \frac{1}{\sin x}\right)$$

Example 9:

$$\lim_{x \to 0^+} \left(\frac{3x-1}{x} - \frac{1}{\sin x}\right)$$

Example 10: Final combo example:

$$\lim_{x \to 0^+} (\sin x)^{1/\ln x}.$$

Suggested after class problems from APEX: Section 6.7: 11,24,35,39

4.3 - Optimization

Goals: Optimize the value of a quantity in a word problem
Before Class Videos: 4.3.1, 4.3.2
Note: This file is labeled Oct 30th, but we will likely only just start this worksheet on Friday, and will do most of it Nov 3rd

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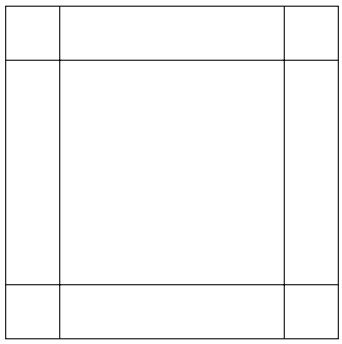
Example 1: A farmer wants to build a rectangular enclosure along a river and has 500 m of fencing for the other three sides. What is the maximum area of the enclosure?

Optimization Steps

1. Draw a diagram. Label relevant parts with variable names.
2. Create an optimization equation involving the variable you want to optimize.
3. Create constraint equations between variables.
4. Use constraint equations to write the optimization equation in terms of one variable.
5. Find (and verify!) extrema.
6. Answer problem.

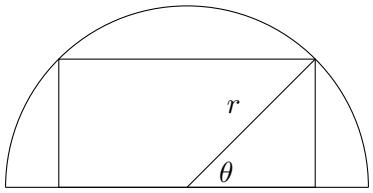
Example 2: An open-top rectangular container has volume 10 m^3. Its length is twice its width. The base material costs \$20/m^2 and the sides cost \$9/m^2. What dimensions minimize the cost?

Example 3: An open-top box is made from a square of side length 8 by cutting squares of side length x from the corners and folding up the resulting flaps. What value of x maximizes the resulting volume of the box.



Example 4: Find the point on $y = \sqrt{x}$ closest to $(3, 0)$.

Example 5: A rectangle is inscribed in a semicircle of radius r.



Example 6: A wire of length L is cut into two at a point x. The left portion is formed into a square. The right portion is formed into a circle. Compute the value of x that maximizes the enclosed area.

Example 7: You went for a swim and need to return to the campground. It is 200 m to swim directly to shore, and then the camp is 1000 m along the shoreline. You can swim at 3 km/h and run at 10 km/h. To what point along the shoreline should you aim so that you minimize the total time to get back to camp?

Suggested after class problems from APEX: Section 4.3: 4, 7, 13, 15

4.1 - Newton's Method

Goals: Apply Newton's method to estimate solutions to equations
Before Class Videos: No video; read Section 4.1 of APEX Calculus 1.
Note: We will use the following DESMOS extensively: `https://www.desmos.com/calculator/y1gjz9bjwn`

Example 1: Recall how IVT implies $f(x) = x^3 - x - 1$ has a root in $(0, 2)$.

Can I use my knowledge of what happens around $x = 1$ to estimate where the root is?

Newton's Method: Goal: $f(x) = 0$. Method: Starting at x_0 estimate a new location x_1 such that $f(x_0) + f'(x_0)(x_1 - x_0) = 0$. We continue iteratively via

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}.$$

Example 2: $x^4 - 2 = 0$ and start with $x_0 = 1$.

Example 3:

$$f(x) = \begin{cases} -\sqrt{1-x}, & x < 1, \\ \sqrt{x-1}, & x \geq 1. \end{cases}$$

With $x_0 = 2$,

Example 4: $\sqrt[3]{x} = 0$ starting at $x_0 = 1$.

Example 5: Estimate where $\cos(x) = 0$. Start $x_0 = 1$.

Suggested after class problems from APEX: Section 4.1: 3, 9, 15

Linearization and Differentials

Goals: Approximate changes using Linearizations and the language of differentials
Before Class Videos: 4.4.1

We can approximate functions by tangent lines. These are numerically easier to compute, and often good enough!

Definition:
The Linearization of $f(x)$ at the point $x = a$ is

$$\boxed{L(x) = f(a) + f'(a)(x - a)}.$$

Example 1: Estimate $\frac{1}{4.1}$ by computing the linearization at $x = 4$.

Now approximate about $x = 1$ instead:

Example 2: Use a Linearization about $x = 0$ to approximate $\frac{1}{0.1}$:

Two lessons about Linearization:

1. Approximations are typically better nearby the point where the linearization is taken.
2. The function must be differentiable at the point to even be computed.

Definition:
Interpret the differential dx as a small change in x. Then the differential $dy = f'(x)dx$ is the resulting small change in y according to the linear approximation at x.

Big Idea:

$$f(a + dx) \approx f(a) + dy.$$

Example 3: Approximate $(7.99)^{1/3}$ using the language of differentials.

Example 4: Approximate the change in area of a circle going from radius $r = 10$ to radius $r = 0.1$.

Suggested after class problems from APEX: Section 4.4: 9, 13, 17, 25, 31, 33